

Boris Benchmarks for Static $E \times B$, Nonuniform $E \times B$, and Polarization Drift

1. Purpose of the benchmark suite

This benchmark suite tests whether a charged-particle pusher correctly reproduces basic drift physics in simple electromagnetic fields. The three cases are deliberately chosen so that the numerical result can be compared directly with analytical theory.

The code advances a 14 keV deuteron using a Boris pusher in Cartesian coordinates. The particle state is stored as

$$\mathbf{x} = (x, y, z), \mathbf{v} = (v_x, v_y, v_z),$$

and the electromagnetic fields are represented as

$$\mathbf{E} = (E_x, E_y, E_z), \mathbf{B} = (B_x, B_y, B_z).$$

Although the pusher is fully three-dimensional, the first two tests are physically two-dimensional because the motion remains in the x - y plane. The third test includes free parallel motion along z , but the drift comparison is still in the transverse x - y plane.

The particle is a deuteron with charge

$$q = e = 1.602176634 \times 10^{-19} \text{ C} \quad ,$$

and mass

$$m_D = 3.343583719 \times 10^{-27} \text{ kg} \quad .$$

The kinetic energy scale is

$$E_{\text{kin}} = 14 \text{ keV} \quad .$$

The corresponding nonrelativistic deuteron speed is

$$v_{14} = \sqrt{\frac{2E_{\text{kin}}}{m_D}} \approx 1.158 \times 10^6 \text{ m} \quad /s.$$

The magnetic field used in all three cases is

$$\mathbf{B} = B_0 \hat{z},$$

with

$$B_0 = 0.5 \text{ T}$$

The deuteron cyclotron frequency is

$$\Omega = \frac{qB_0}{m_D},$$

which gives

$$\Omega \approx 2.396 \times 10^7 \text{ rad/s}.$$

The gyroperiod is

$$T_{\text{gyro}} = \frac{2\pi}{\Omega} \approx 2.622 \times 10^{-7} \text{ s}.$$

The timestep is

$$\Delta t = 2.0 \times 10^{-10} \text{ s},$$

so that

$$\frac{\Delta t}{T_{\text{gyro}}} \approx 7.63 \times 10^{-4}.$$

This timestep is small enough that the drift comparisons are not dominated by cyclotron phase error.

2. Numerical method

The code uses a leapfrog-consistent Boris pusher. Positions are stored at integer times,

$$\mathbf{x}_n \equiv \mathbf{x}(t_n),$$

while velocities are advanced naturally at half times,

$$\mathbf{v}_{n+1/2}.$$

This distinction is important. A standard Boris method is time-centered, so using integer-time positions together with improperly centered velocities can produce an artificial offset in guiding-center diagnostics. In particular, for the static $E \times B$ test, that mismatch can make the guiding center appear to drift in x , even though theory says it should remain fixed.

For each timestep, the velocity is advanced from $\mathbf{v}_{n-1/2}$ to $\mathbf{v}_{n+1/2}$ using the usual Boris sequence. With electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$, the first half electric kick is

$$\mathbf{v}^- = \mathbf{v}_{n-1/2} + \frac{q\mathbf{E} \Delta t}{m} \frac{1}{2}.$$

The magnetic rotation is written using

$$\begin{aligned} \mathbf{t} &= \frac{q\mathbf{B} \Delta t}{m} \frac{1}{2}, \\ \mathbf{s} &= \frac{2\mathbf{t}}{1 + t^2}. \end{aligned}$$

Then

$$\begin{aligned} \mathbf{v}' &= \mathbf{v}^- + \mathbf{v}^- \times \mathbf{t}, \\ \mathbf{v}^+ &= \mathbf{v}^- + \mathbf{v}' \times \mathbf{s}. \end{aligned}$$

The second half electric kick is

$$\mathbf{v}_{n+1/2} = \mathbf{v}^+ + \frac{q\mathbf{E} \Delta t}{m} \frac{1}{2}.$$

The position is updated by

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{v}_{n+1/2} \Delta t.$$

For diagnostics, the code reconstructs a centered velocity at integer time, $\mathbf{v}_n$, by advancing the half-step velocity over a half timestep. This is the key correction that makes the guiding-center comparisons clean.

3. Case 1: Uniform static $E \times B$ drift

3.1 Field configuration

The first test uses uniform, static fields:

$$\begin{aligned}\mathbf{E} &= E_0 \hat{x}, \\ \mathbf{B} &= B_0 \hat{z}.\end{aligned}$$

The electric field strength is

$$E_0 = 2.0 \times 10^4 \text{ V} \quad \text{/m.}$$

The theoretical $E \times B$ drift is

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2}.$$

Since

$$\mathbf{E} = E_0 \hat{x}, \mathbf{B} = B_0 \hat{z},$$

we have

$$\mathbf{E} \times \mathbf{B} = E_0 B_0 (\hat{x} \times \hat{z}) = -E_0 B_0 \hat{y}.$$

Therefore,

$$\mathbf{v}_E = -\frac{E_0}{B_0} \hat{y}.$$

For the parameters used here,

$$v_{E,y} = -\frac{2.0 \times 10^4}{0.5} = -4.0 \times 10^4 \text{ m/s.}$$

3.2 Exact particle solution

The particle velocity can be decomposed into drift plus gyromotion:

$$\mathbf{v}(t) = \mathbf{v}_E + \mathbf{u}(t),$$

where $\mathbf{u}(t)$ is the cyclotron velocity in the drifting frame.

For a positive ion in $\mathbf{B} = B_0 \hat{z}$, the cyclotron velocity satisfies

$$\frac{d\mathbf{u}}{dt} = \Omega \mathbf{u} \times \hat{z}.$$

Writing the initial cyclotron velocity as

$$\mathbf{u}_0 = (u_{0x}, u_{0y}, 0),$$

the exact solution is

$$\begin{aligned} u_x(t) &= u_{0x} \cos(\Omega t) + u_{0y} \sin(\Omega t), \\ u_y(t) &= -u_{0x} \sin(\Omega t) + u_{0y} \cos(\Omega t). \end{aligned}$$

The velocity is therefore

$$\begin{aligned} v_x(t) &= u_x(t), \\ v_y(t) &= u_y(t) + v_{E,y}. \end{aligned}$$

The position is obtained by integrating velocity:

$$\begin{aligned} x(t) &= x_0 + \frac{u_{0x}}{\Omega} \sin(\Omega t) + \frac{u_{0y}}{\Omega} [1 - \cos(\Omega t)], \\ y(t) &= y_0 + \frac{u_{0x}}{\Omega} [\cos(\Omega t) - 1] + \frac{u_{0y}}{\Omega} \sin(\Omega t) + v_{E,y} t. \end{aligned}$$

The guiding center is

$$\mathbf{R}_{gc} = \mathbf{x} + \frac{(\mathbf{v} - \mathbf{v}_E) \times \hat{z}}{\Omega}.$$

For this uniform case, the exact guiding-center motion is simply

$$\begin{aligned} X_{gc}(t) &= X_{gc}(0), \\ Y_{gc}(t) &= Y_{gc}(0) + v_{E,y}t. \end{aligned}$$

Thus, the guiding center should drift only in $-y$, with no secular drift in x .

3.3 Numerical result

The numerical result agrees very closely with theory. The run produced

$$v_{E,y} = -4.0 \times 10^4 \text{ m/s},$$

with a 14 keV Larmor radius

$$\rho_L = \frac{v_{\perp}}{\Omega} \approx 4.83 \times 10^{-2} \text{ m}.$$

The corrected leapfrog-consistent diagnostic gives

$$\begin{aligned} \text{RMS}(X_{gc} - X_{gc,\text{exact}}) &\approx 9.81 \times 10^{-8} \text{ m}, \\ \text{RMS}(Y_{gc} - Y_{gc,\text{exact}}) &\approx 1.70 \times 10^{-7} \text{ m}. \end{aligned}$$

The final guiding-center errors are

$$\begin{aligned} \Delta X_{gc,\text{final}} &\approx 1.76 \times 10^{-10} \text{ m}, \\ \Delta Y_{gc,\text{final}} &\approx 8.99 \times 10^{-14} \text{ m}. \end{aligned}$$

This confirms that the numerical pusher reproduces the static $E \times B$ drift and that the earlier artificial x -drift was a diagnostic centering issue, not a physical or numerical drift of the Boris orbit.

4. Case 2: Strong-gradient nonuniform static $E \times B$ -like motion

4.1 Field configuration

The second test uses a spatially varying but static electric field,

$$\mathbf{E} = E_x(x)\hat{x},$$

with

$$E_x(x) = E_0 + Gx.$$

The magnetic field remains uniform:

$$\mathbf{B} = B_0\hat{z}.$$

Here,

$$E_0 = 2.0 \times 10^4 \text{ V/m},$$

and the gradient is

$$G = \frac{dE_x}{dx}.$$

In the strong-gradient version,

$$G = 1.198 \times 10^6 \text{ V/m}^2.$$

It is useful to define

$$\gamma = \frac{qG}{m_D}.$$

The quantity γ has units of s^{-2} . It measures the strength of the linear electric-field force relative to particle inertia. The dimensionless gradient strength is

$$\frac{\gamma}{\Omega^2}.$$

In this test,

$$\frac{\gamma}{\Omega^2} = 0.10.$$

This value is strong enough to make the nonuniform drift visibly different from the uniform drift, while still keeping the motion stable and analytically tractable.

4.2 Equations of motion

The Lorentz force equations are

$$\begin{aligned}\frac{dv_x}{dt} &= \frac{q}{m_D} (E_0 + Gx) + \Omega v_y, \\ \frac{dv_y}{dt} &= -\Omega v_x.\end{aligned}$$

The second equation implies

$$\frac{d}{dt}(v_y + \Omega x) = 0.$$

Therefore,

$$v_y + \Omega x = C,$$

where C is a constant set by the initial condition.

Using

$$v_y = C - \Omega x,$$

in the x -equation gives

$$\frac{d^2 x}{dt^2} = \frac{qE_0}{m_D} + \frac{qG}{m_D} x + \Omega(C - \Omega x).$$

Using $\gamma = qG/m_D$, this becomes

$$\frac{d^2x}{dt^2} + (\Omega^2 - \gamma)x = \frac{qE_0}{m_D} + \Omega C.$$

Define the modified gyrofrequency

$$\kappa = \sqrt{\Omega^2 - \gamma}.$$

For stable oscillatory motion, we need

$$\gamma < \Omega^2.$$

The code chooses an oscillation center x_{eq} , and sets the constant C so that $x = x_{\text{eq}}$ is the equilibrium position. The equilibrium condition is

$$(\Omega^2 - \gamma)x_{\text{eq}} = \frac{qE_0}{m_D} + \Omega C.$$

Solving for C ,

$$C = \frac{(\Omega^2 - \gamma)x_{\text{eq}} - (qE_0/m_D)}{\Omega}.$$

The exact solution is then

$$\begin{aligned} x(t) &= x_{\text{eq}} + A \cos(\kappa t), \\ v_x(t) &= -A \kappa \sin(\kappa t). \end{aligned}$$

Since

$$v_y = C - \Omega x,$$

we get

$$v_y(t) = C - \Omega x_{\text{eq}} - \Omega A \cos(\kappa t).$$

Define the mean drift velocity

$$\bar{v}_y = C - \Omega x_{\text{eq}}.$$

Then

$$v_y(t) = \bar{v}_y - \Omega A \cos(\kappa t).$$

The corresponding position in y is

$$y(t) = y_0 + \bar{v}_y t - \frac{\Omega A}{\kappa} \sin(\kappa t).$$

The mean drift velocity can also be written as the local $E \times B$ drift at the oscillation center:

$$\bar{v}_y = -\frac{E_0 + G x_{\text{eq}}}{B_0}.$$

Thus, for a linear electric field, the particle has an exact modified cyclotron orbit whose average transverse drift equals the local $E \times B$ drift evaluated at x_{eq} .

4.3 Parameters

In the strong-gradient run,

$$\frac{\gamma}{\Omega^2} = 0.10,$$

so

$$\kappa = \Omega \sqrt{1 - 0.10} = 0.9487 \Omega.$$

The chosen oscillation center is

$$x_{\text{eq}} = 0.30 \text{ m}.$$

The oscillation amplitude is

$$A \approx 5.10 \times 10^{-2} \text{ m} \quad .$$

The local electric field at the oscillation center is

$$E_x(x_{\text{eq}}) = E_0 + Gx_{\text{eq}}.$$

With

$$\begin{aligned} E_0 &= 2.0 \times 10^4 \text{ V} & / \text{m}, \\ G &= 1.198 \times 10^6 \text{ V} & / \text{m}^2, \\ x_{\text{eq}} &= 0.30 \text{ m} & , \end{aligned}$$

we get

$$\begin{aligned} E_x(x_{\text{eq}}) &\approx 2.0 \times 10^4 + (1.198 \times 10^6)(0.30) \\ &\approx 3.79 \times 10^5 \text{ V} & / \text{m}. \end{aligned}$$

Thus,

$$\bar{v}_y = -\frac{E_x(x_{\text{eq}})}{B_0} \approx -7.59 \times 10^5 \text{ m} \quad / \text{s}.$$

This is much larger than the uniform Case 1 drift,

$$v_{E,y}^{\text{uniform}} = -4.0 \times 10^4 \text{ m} \quad / \text{s}.$$

So the gradient case is about

$$\frac{7.59 \times 10^5}{4.0 \times 10^4} \approx 19$$

times faster in mean drift speed.

4.4 Numerical result

The code compares the numerical Boris result to the exact linear-field orbit. The run produced

$$\bar{v}_{y,\text{exact}} = -7.587692 \times 10^5 \text{ m/s},$$

and

$$\bar{v}_{y,\text{numerical}} = -7.587516 \times 10^5 \text{ m/s}.$$

The agreement is excellent. The relative errors were

$$\begin{aligned}\epsilon_x &\approx 1.74 \times 10^{-5}, \\ \epsilon_y &\approx 2.33 \times 10^{-6}, \\ \epsilon_{v_x} &\approx 1.46 \times 10^{-4}, \\ \epsilon_{v_y} &\approx 1.10 \times 10^{-4}.\end{aligned}$$

The final position errors were

$$\begin{aligned}\Delta x_{\text{final}} &\approx -2.12 \times 10^{-8} \text{ m}, \\ \Delta y_{\text{final}} &\approx 1.55 \times 10^{-5} \text{ m}.\end{aligned}$$

This case verifies that the pusher captures nonuniform static electric-field motion and that the observed drift matches the exact linear-field theory.

5. Case 3: Magnetized polarization drift

5.1 Field configuration

The third test uses a time-dependent electric field with a uniform magnetic field:

$$\begin{aligned}\mathbf{E}(t) &= E_{\text{amp}} \sin(\omega t) \hat{x}, \\ \mathbf{B} &= B_0 \hat{z}.\end{aligned}$$

The electric-field amplitude is

$$E_{\text{amp}} = 3.0 \times 10^4 \text{ V/m}.$$

The frequency is

$$f = 2.5 \times 10^5 \text{ Hz} \quad ,$$

so

$$\omega = 2\pi f.$$

The important dimensionless parameter is

$$\epsilon = \frac{\omega}{\Omega}.$$

For this run,

$$\epsilon = \frac{\omega}{\Omega} \approx 6.56 \times 10^{-2}.$$

Since $\epsilon \ll 1$, the first-order polarization drift is expected to be a good approximation, but not exact. The code therefore compares the numerical orbit to the exact finite-frequency driven solution and also plots the first-order drift approximations.

The 14 keV deuteron energy is included as parallel motion along the magnetic field:

$$v_z = v_{14}.$$

This preserves the same particle energy scale while avoiding contamination of the transverse drift comparison. Because the fields are uniform in z , this parallel motion is dynamically decoupled from the transverse motion.

5.2 First-order drift theory

The instantaneous $E \times B$ drift is

$$\mathbf{v}_E(t) = \frac{\mathbf{E}(t) \times \mathbf{B}}{B^2}.$$

For

$$\begin{aligned} \mathbf{E}(t) &= E_{\text{amp}} \sin(\omega t) \hat{x}, \\ \mathbf{B} &= B_0 \hat{z}, \end{aligned}$$

we have

$$v_{E,y}(t) = -\frac{E_{\text{amp}}}{B_0} \sin(\omega t).$$

This drift oscillates in time and has zero average over a complete number of electric-field periods.

The first-order polarization drift is

$$\mathbf{v}_{\text{pol}} = \frac{m_D}{qB_0^2} \frac{d\mathbf{E}_{\perp}}{dt}.$$

Since

$$\begin{aligned} E_x(t) &= E_{\text{amp}} \sin(\omega t), \\ \frac{dE_x}{dt} &= E_{\text{amp}} \omega \cos(\omega t). \end{aligned}$$

Thus,

$$v_{\text{pol},x}(t) = \frac{m_D}{qB_0^2} E_{\text{amp}} \omega \cos(\omega t).$$

Using

$$\Omega = \frac{qB_0}{m_D},$$

this can also be written as

$$v_{\text{pol},x}(t) = \frac{E_{\text{amp}}}{B_0} \frac{\omega}{\Omega} \cos(\omega t).$$

Therefore, the polarization drift velocity is smaller than the oscillatory $E \times B$ velocity by approximately

$$\frac{v_{\text{pol}}}{v_E} \sim \frac{\omega}{\Omega}.$$

For this case,

$$\frac{\omega}{\Omega} \approx 0.0656.$$

So the polarization velocity amplitude is about 6.6% of the oscillatory $E \times B$ velocity amplitude.

5.3 Exact finite-frequency theory

For this special case, the transverse equations of motion can be solved exactly. The equations are

$$\begin{aligned} \frac{dv_x}{dt} &= \frac{q}{m_D} E_{\text{amp}} \sin(\omega t) + \Omega v_y, \\ \frac{dv_y}{dt} &= -\Omega v_x. \end{aligned}$$

For the driven finite-frequency solution, excluding free cyclotron transients, the exact transverse velocities are

$$\begin{aligned} v_x(t) &= \frac{E_{\text{amp}}}{B_0} \frac{\epsilon}{1 - \epsilon^2} \cos(\omega t), \\ v_y(t) &= -\frac{E_{\text{amp}}}{B_0} \frac{1}{1 - \epsilon^2} \sin(\omega t), \end{aligned}$$

where

$$\epsilon = \frac{\omega}{\Omega}.$$

In the low-frequency limit,

$$\epsilon \ll 1,$$

these reduce to

$$v_x(t) \approx \frac{E_{\text{amp}}}{B_0} \epsilon \cos(\omega t),$$

which is the polarization drift, and

$$v_y(t) \approx -\frac{E_{\text{amp}}}{B_0} \sin(\omega t),$$

which is the time-dependent $E(t) \times B$ drift.

The exact transverse positions are obtained by integrating:

$$\begin{aligned} x(t) &= x_0 + \frac{E_{\text{amp}}}{B_0} \frac{\epsilon}{1 - \epsilon^2} \frac{\sin(\omega t)}{\omega}, \\ y(t) &= y_0 + \frac{E_{\text{amp}}}{B_0} \frac{1}{1 - \epsilon^2} \frac{\cos(\omega t) - 1}{\omega}. \end{aligned}$$

The simulation is initialized directly on this exact driven solution at the half-step velocity. This removes startup cyclotron transients and avoids the beginning/end offset problem seen in earlier versions.

5.4 Numerical result

The run used

$$\begin{aligned} E_{\text{amp}} &= 3.0 \times 10^4 \text{ V} & / \text{m}, \\ f &= 2.5 \times 10^5 \text{ Hz} & , \\ \epsilon &= \frac{\omega}{\Omega} \approx 6.56 \times 10^{-2}. \end{aligned}$$

The comparison against the exact finite-frequency solution produced

$$\begin{aligned} \epsilon_x &\approx 2.22 \times 10^{-4}, \\ \epsilon_y &\approx 9.09 \times 10^{-5}, \\ \epsilon_{v_x} &\approx 2.40 \times 10^{-3}, \\ \epsilon_{v_y} &\approx 2.22 \times 10^{-4}. \end{aligned}$$

The first-order drift approximations differ from the exact finite-frequency solution by

$$\epsilon_{\text{pol,first}} \approx 4.30 \times 10^{-3},$$

and

$$\epsilon_{E \times B, first} \approx 4.30 \times 10^{-3}.$$

This is consistent with the expected finite-frequency correction of order

$$\epsilon^2 = \left(\frac{\omega}{\Omega}\right)^2 \approx 4.30 \times 10^{-3}.$$

The final position errors were

$$\begin{aligned} \Delta x_{\text{final}} &\approx 3.70 \times 10^{-9} \text{ m} & , \\ \Delta y_{\text{final}} &\approx 5.40 \times 10^{-8} \text{ m} & , \\ \Delta z_{\text{final}} &\approx -1.86 \times 10^{-11} \text{ m} & . \end{aligned}$$

Thus, the numerical result agrees closely with the exact finite-frequency theory, while the first-order polarization drift shows the expected small finite- ω/Ω deviation.

6. Summary of the three tests

The three tests verify progressively more complex drift physics.

The first test verifies the canonical static $E \times B$ drift in uniform fields. The result is a constant guiding-center drift,

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2},$$

with no spurious drift in the transverse direction. The corrected leapfrog-consistent diagnostic makes the guiding-center x -coordinate remain fixed to within numerical precision.

The second test verifies motion in a nonuniform static electric field,

$$E_x(x) = E_0 + Gx.$$

Because the field is linear in x , the exact orbit is known. The particle undergoes modified cyclotron motion with frequency

$$\kappa = \sqrt{\Omega^2 - \gamma},$$

where

$$\gamma = \frac{qG}{m_D}.$$

The mean drift is the local $E \times B$ drift evaluated at the oscillation center:

$$\bar{v}_y = -\frac{E_0 + Gx_{\text{eq}}}{B_0}.$$

In the strong-gradient case, this drift is about 19 times larger than the uniform-drift case, making the distinction visually clear.

The third test verifies magnetized polarization drift in a time-dependent electric field,

$$E_x(t) = E_{\text{amp}} \sin(\omega t).$$

The first-order polarization drift is

$$v_{\text{pol},x} = \frac{m_D}{qB_0^2} \frac{dE_x}{dt},$$

and the accompanying time-dependent $E \times B$ drift is

$$v_{E,y} = -\frac{E_x(t)}{B_0}.$$

The code compares the numerical orbit to the exact finite-frequency solution, showing that the Boris pusher reproduces both the polarization motion and the oscillatory $E(t) \times B$ motion accurately.

Overall, this benchmark suite demonstrates that the corrected leapfrog-consistent Boris implementation captures static $E \times B$ drift, nonuniform static electric-field drift, and polarization drift with clean agreement to analytical theory.